Course 3031 | Semester III | Theory | 4 Credits

Electrical and Electronic Measuring Instruments

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 3031 as **Electrical and Electronic Measuring Instruments** in Semester III for Electrical & Electronics Engineering. The official SITTTR programme/detail page could not be retrieved during this cycle. With user approval, explanatory coverage is based on reliable public electrical-measurement curriculum sources and is not presented as recovered official Course 3031 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment entries are provisional handbook placeholders because the official SITTTR source could not be accessed. The topic sequence and study support are drawn from cited public curricula on electrical/electronic measurement. Reconcile all assessment and exact-topic requirements with official material when available.

> **Verified source note:** Public curriculum sources were retrieved successfully and are linked in References. The official SITTTR source remained inaccessible during the automated cycle.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

This course develops measurement literacy for electrical systems: instrument characteristics and errors, analog meters and transformers, power/energy measurement and bridges, followed by electronic instruments, oscilloscopes and transducers.

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Prerequisite foundation

Fundamentals of mathematics, DC/AC circuits and basic electronic devices.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain measurement-system terms, instrument characteristics and errors.
2. Describe analog instruments, range extension and instrument transformers.
3. Apply methods for power, energy, resistance, inductance and capacitance measurement.
4. Explain electronic instruments, CRO/DSO operation and sensor/transducer selection.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain measurement characteristics, errors and operating principles of electrical instruments.	Understand
CO2	Select and interpret analog instruments and instrument transformers for an electrical measurement task.	Apply
CO3	Explain methods for measuring power, energy and electrical quantities using bridges and potentiometers.	Apply
CO4	Explain electronic measurement with oscilloscopes, digital instruments and transducers.	Apply

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Measurement Fundamentals and Analog Instruments	Measurement terms, static characteristics, errors, PMMC/MI/electrodynamometer/induction/electrostatic instruments and range extension.	Approx. 16 hours	CO1	Module 1
2	Instrument Transformers, Power and Energy Measurement	CT/PT principles and errors; wattmeters; single/three-phase power; induction energy meters and testing.	Approx. 15 hours	CO2	Module 2
3	Potentiometers and Bridge Measurements	DC/AC potentiometers; low/medium/high resistance; Wheatstone/Kelvin bridges; inductance/capacitance/frequency bridges.	Approx. 15 hours	CO3	Module 3
4	Electronic Instruments, Oscilloscopes and Transducers	CRO/DSO operation; voltage/frequency/phase measurement; digital meters; signal sources; sensors, LVDT, strain gauge and temperature/flow transducers.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Measurement Fundamentals and Analog Instruments

Measurement terms, static characteristics, errors, PMMC/MI/electrodynamometer/induction/electrostatic instruments and range extension.

CO1 · Approx. 16 hours

Instrument Transformers, Power and Energy Measurement

CT/PT principles and errors; wattmeters; single/three-phase power; induction energy meters and testing.

CO2 · Approx. 15 hours

Potentiometers and Bridge Measurements

DC/AC potentiometers; low/medium/high resistance; Wheatstone/Kelvin bridges; inductance/capacitance/frequency bridges.

CO3 · Approx. 15 hours

Electronic Instruments, Oscilloscopes and Transducers

CRO/DSO operation; voltage/frequency/phase measurement; digital meters; signal sources; sensors, LVDT, strain gauge and temperature/flow transducers.

CO4 · Approx. 14 hours

MODULE 1

Measurement Fundamentals and Analog Instruments

Measurement terms, static characteristics, errors, PMMC/MI/electrodynamometer/induction/electrostatic instruments and range extension.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

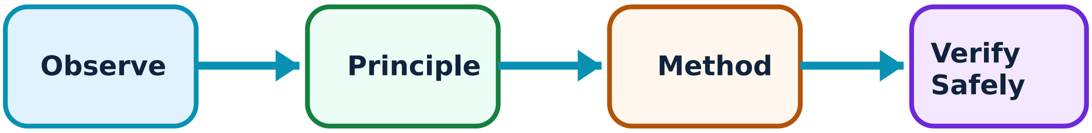
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Measurement Fundamentals and Analog Instruments: identify the system, state its principle, follow the correct method and verify the result safely.

1. Measurement systems, characteristics and errors−

Accuracy, precision, resolution, sensitivity and speed of response describe measurement quality. Gross, systematic and random errors, plus loading effects, influence a reported value and its reliability.

Study and application method

State measurand, instrument range/resolution and likely error sources; distinguish repeatability from closeness to reference value before selecting a method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State measurand, instrument range/resolution and likely error sources; distinguish repeatability from closeness to reference value before selecting a method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A precise reading can be inaccurate when systematic error or calibration offset is present.

2. PMMC, moving-iron and electrodynamometer instruments−

Analog instruments convert electrical quantity into deflecting torque balanced by control torque. PMMC, moving-iron and electrodynamometer mechanisms differ in current type, scale and application.

Study and application method

Use construction → operating principle → torque/scale → application → limitation sequence and include a labelled movement sketch.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use construction → operating principle → torque/scale → application → limitation sequence and include a labelled movement sketch.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use a DC-only PMMC movement directly for AC measurement without the specified rectification arrangement.

3. Range extension and electrostatic instruments−

Shunts extend current ranges and series multipliers extend voltage ranges. Electrostatic voltmeters use electrostatic attraction and suit high-voltage measurement under appropriate conditions.

Study and application method

Identify desired range, meter full-scale data and connection type, calculate the added element and check power/rating requirement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect shunt/multiplier connection can damage a sensitive movement; confirm polarity and rating before energising.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Instrument Transformers, Power and Energy Measurement

CT/PT principles and errors; wattmeters; single/three-phase power; induction energy meters and testing.

Approx. 15 hours **CO2** **Understand · Apply**

Learning goals

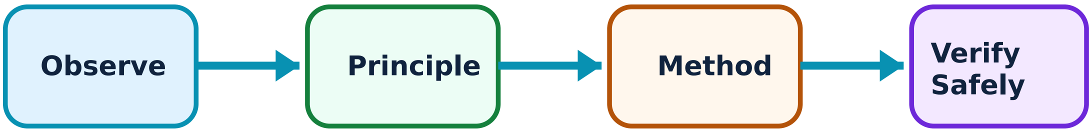
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Instrument Transformers, Power and Energy Measurement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Current and potential transformers−

Instrument transformers scale high current or voltage to safe standard values for meters and relays. Ratio and phase-angle errors affect measurement, especially at different burden and power-factor conditions.

Study and application method

Identify primary/secondary ratings, instrument burden and safety requirement; state connection and ratio before calculating the primary quantity.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify primary/secondary ratings, instrument burden and safety requirement; state connection and ratio before calculating the primary quantity.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never leave a current-transformer secondary open while its primary is energised.

2. Wattmeters and power measurement−

Electrodynamometer and induction-type principles support power measurement. Single- and three-phase methods depend on connection, load balance and required active/reactive power quantity.

Study and application method

Draw current and pressure coil connections, mark polarities, identify supply/load configuration and state a cross-check for plausibility.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw current and pressure coil connections, mark polarities, identify supply/load configuration and state a cross-check for plausibility.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A wattmeter reading can be negative or incorrect with reversed connections; do not change wiring under live conditions.

3. Energy meters−

An energy meter accumulates electrical energy over time. Induction-type meters use driving and braking torques; digital meters sample and calculate energy electronically. Testing compares meter registration with a known reference.

Study and application method

Record initial/final register, load and elapsed time, calculate reference energy and determine percentage error with stated sign convention.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Record initial/final register, load and elapsed time, calculate reference energy and determine percentage error with stated sign convention.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse power (kW) with energy (kWh).

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Potentiometers and Bridge Measurements

DC/AC potentiometers; low/medium/high resistance; Wheatstone/Kelvin bridges; inductance/capacitance/frequency bridges.

Approx. 15 hours | CO3 | Understand · Apply

Learning goals

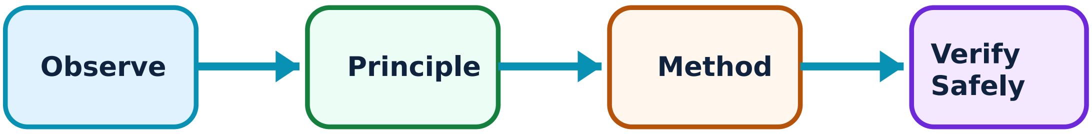
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Potentiometers and Bridge Measurements: identify the system, state its principle, follow the correct method and verify the result safely.

1. Potentiometers−

A potentiometer compares an unknown voltage with a known calibrated voltage under null-balance condition, minimising loading of the source. DC and AC forms support different quantities and phase relationships.

Study and application method

Explain standardisation first, show null-detection connection, then state how the balance relation yields the unknown quantity.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain standardisation first, show null-detection connection, then state how the balance relation yields the unknown quantity.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A null method requires correct reference adjustment; do not read an unknown before standardisation.

2. Resistance measurement and DC bridges−

Medium resistance is commonly measured through bridge comparison; Kelvin methods reduce lead/contact influence for low resistance; high-resistance methods use appropriate insulation/leakage precautions.

Study and application method

Select method by resistance range, draw bridge arms/unknown, write balance condition and state one dominant error control.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Select method by resistance range, draw bridge arms/unknown, write balance condition and state one dominant error control.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Lead/contact resistance can dominate a low-resistance measurement; do not use a two-wire method blindly.

3. AC bridges−

AC bridges compare impedance to determine inductance, capacitance, resistance or frequency. Balance involves both magnitude and phase components, so component selection and frequency condition matter.

Study and application method

Name bridge, state measured component, draw balance arms and write the balance relation only after identifying the circuit’s intended condition.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Name bridge, state measured component, draw balance arms and write the balance relation only after identifying the circuit’s intended condition.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not substitute DC-resistance logic into an AC bridge without treating reactance and frequency.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Electronic Instruments, Oscilloscopes and Transducers

CRO/DSO operation; voltage/frequency/phase measurement; digital meters; signal sources; sensors, LVDT, strain gauge and temperature/flow transducers.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

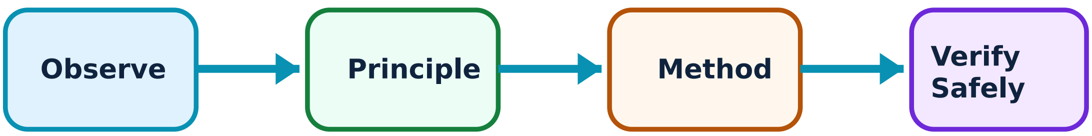
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electronic Instruments, Oscilloscopes and Transducers: identify the system, state its principle, follow the correct method and verify the result safely.

1. CRO and digital storage oscilloscope−

Oscilloscopes display voltage versus time and support amplitude, time, frequency and phase interpretation. CRO subsystems include vertical/horizontal amplification, triggering and time base; DSO adds sampling/storage.

Study and application method

Check probe attenuation and ground reference, set time/div and volts/div, obtain a stable trigger, then read scale values with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Check probe attenuation and ground reference, set time/div and volts/div, obtain a stable trigger, then read scale values with units.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Instrument ground connection can create a fault in non-isolated circuits; follow safe probing practice.

2. Digital instruments and signal generators−

Digital voltmeters, multimeters and frequency meters provide numerical readings with specified resolution, sensitivity and input characteristics. Signal generators create controlled test waveforms.

Study and application method

Select correct function/range, verify input rating/category, connect safely and record display resolution/uncertainty with the reading.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Select correct function/range, verify input rating/category, connect safely and record display resolution/uncertainty with the reading.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: [illegible] concept [illegible]. [illegible] diagram / circuit / formula / procedure [illegible]. [illegible] result [illegible] application [illegible]. Technical terms English-[illegible] [illegible].

Common mistake / precaution: A digital display is not automatically more accurate; range, calibration and input loading still matter.

3. Sensors and transducers−

A transducer converts a physical quantity into a usable signal. LVDT, strain gauges, temperature sensors and magnetic-flow systems have distinct principles, conditioning needs and applications.

Study and application method

Identify measurand, range, environment, sensitivity, output type and required conditioning; then choose a transducer with a clear justification.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Do not select a sensor only by nominal range; temperature, vibration, isolation and calibration requirements matter.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Measurement Fundamentals and Analog Instruments

Use the concept flow to revise observation, principle, method and verification.

Module 2: Instrument Transformers, Power and Energy Measurement

Use the concept flow to revise observation, principle, method and verification.

Module 3: Potentiometers and Bridge Measurements

Use the concept flow to revise observation, principle, method and verification.

Module 4: Electronic Instruments, Oscilloscopes and Transducers

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create an error-and-calibration table for a selected meter and explain accuracy versus precision.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw safe CT/PT and wattmeter connections, including one key safety precaution.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Solve a bridge/potentiometer balance exercise and identify the measurement range for which the method is appropriate.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Capture a low-voltage waveform safely with an oscilloscope and compare two candidate transducers for an industrial measurand.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Measurement Fundamentals and Analog Instruments

Explain Measurement systems, characteristics and errors and state one correct method, application or precaution.−

Accuracy, precision, resolution, sensitivity and speed of response describe measurement quality. Gross, systematic and random errors, plus loading effects, influence a reported value and its reliability.

Answer framework: State measurand, instrument range/resolution and likely error sources; distinguish repeatability from closeness to reference value before selecting a method.

Important point: A precise reading can be inaccurate when systematic error or calibration offset is present.

Explain PMMC, moving-iron and electrodynamometer instruments and state one correct method, application or precaution.−

Analog instruments convert electrical quantity into deflecting torque balanced by control torque. PMMC, moving-iron and electrodynamometer mechanisms differ in current type, scale and application.

Answer framework: Use construction → operating principle → torque/scale → application → limitation sequence and include a labelled movement sketch.

Important point: Do not use a DC-only PMMC movement directly for AC measurement without the specified rectification arrangement.

Explain Range extension and electrostatic instruments and state one correct method, application or precaution.−

Shunts extend current ranges and series multipliers extend voltage ranges. Electrostatic voltmeters use electrostatic attraction and suit high-voltage measurement under appropriate conditions.

Answer framework: Identify desired range, meter full-scale data and connection type, calculate the added element and check power/rating requirement.

Important point: Incorrect shunt/multiplier connection can damage a sensitive movement; confirm polarity and rating before energising.

Module 2 — Instrument Transformers, Power and Energy Measurement

Explain Current and potential transformers and state one correct method, application or precaution.−

Instrument transformers scale high current or voltage to safe standard values for meters and relays. Ratio and phase-angle errors affect measurement, especially at different burden and power-factor conditions.

Answer framework: Identify primary/secondary ratings, instrument burden and safety requirement; state connection and ratio before calculating the primary quantity.

Important point: Never leave a current-transformer secondary open while its primary is energised.

Explain Wattmeters and power measurement and state one correct method, application or precaution.−

Electrodynamometer and induction-type principles support power measurement. Single- and three-phase methods depend on connection, load balance and required active/reactive power quantity.

Answer framework: Draw current and pressure coil connections, mark polarities, identify supply/load configuration and state a cross-check for plausibility.

Important point: A wattmeter reading can be negative or incorrect with reversed connections; do not change wiring under live conditions.

Explain Energy meters and state one correct method, application or precaution.−

An energy meter accumulates electrical energy over time. Induction-type meters use driving and braking torques; digital meters sample and calculate energy electronically. Testing compares meter registration with a known reference.

Answer framework: Record initial/final register, load and elapsed time, calculate reference energy and determine percentage error with stated sign convention.

Important point: Do not confuse power (kW) with energy (kWh).

Module 3 — Potentiometers and Bridge Measurements

Explain Potentiometers and state one correct method, application or precaution.−

A potentiometer compares an unknown voltage with a known calibrated voltage under null-balance condition, minimising loading of the source. DC and AC forms support different quantities and phase relationships.

Answer framework: Explain standardisation first, show null-detection connection, then state how the balance relation yields the unknown quantity.

Important point: A null method requires correct reference adjustment; do not read an unknown before standardisation.

Explain Resistance measurement and DC bridges and state one correct method, application or precaution.−

Medium resistance is commonly measured through bridge comparison; Kelvin methods reduce lead/contact influence for low resistance; high-resistance methods use appropriate insulation/leakage precautions.

Answer framework: Select method by resistance range, draw bridge arms/unknown, write balance condition and state one dominant error control.

Important point: Lead/contact resistance can dominate a low-resistance measurement; do not use a two-wire method blindly.

Explain AC bridges and state one correct method, application or precaution.−

AC bridges compare impedance to determine inductance, capacitance, resistance or frequency. Balance involves both magnitude and phase components, so component selection and frequency condition matter.

Answer framework: Name bridge, state measured component, draw balance arms and write the balance relation only after identifying the circuit's intended condition.

Important point: Do not substitute DC-resistance logic into an AC bridge without treating reactance and frequency.

Module 4 — Electronic Instruments, Oscilloscopes and Transducers

Explain CRO and digital storage oscilloscope and state one correct method, application or precaution.−

Oscilloscopes display voltage versus time and support amplitude, time, frequency and phase interpretation. CRO subsystems include vertical/horizontal amplification, triggering and time base; DSO adds sampling/storage.

Answer framework: Check probe attenuation and ground reference, set time/div and volts/div, obtain a stable trigger, then read scale values with units.

Important point: Instrument ground connection can create a fault in non-isolated circuits; follow safe probing practice.

Explain Digital instruments and signal generators and state one correct method, application or precaution.−

Digital voltmeters, multimeters and frequency meters provide numerical readings with specified resolution, sensitivity and input characteristics. Signal generators create controlled test waveforms.

Answer framework: Select correct function/range, verify input rating/category, connect safely and record display resolution/uncertainty with the reading.

Important point: A digital display is not automatically more accurate; range, calibration and input loading still matter.

Explain Sensors and transducers and state one correct method, application or precaution.−

A transducer converts a physical quantity into a usable signal. LVDT, strain gauges, temperature sensors and magnetic-flow systems have distinct principles, conditioning needs and applications.

Answer framework: Identify measurand, range, environment, sensitivity, output type and required conditioning; then choose a transducer with a clear justification.

Important point: Do not select a sensor only by nominal range; temperature, vibration, isolation and calibration requirements matter.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Measurement Fundamentals and Analog Instruments.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Instrument Transformers, Power and Energy Measurement.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Potentiometers and Bridge Measurements.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Electronic Instruments, Oscilloscopes and Transducers.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Measurement terms, static characteristics, errors, PMMC/MI/electrodynamometer/induction/electrostatic instruments and range extension..
2. Develop a complete answer or practical plan for: CT/PT principles and errors; wattmeters; single/three-phase power; induction energy meters and testing..
3. Develop a complete answer or practical plan for: DC/AC potentiometers; low/medium/high resistance; Wheatstone/Kelvin bridges; inductance/capacitance/frequency bridges..
4. Develop a complete answer or practical plan for: CRO/DSO operation; voltage/frequency/phase measurement; digital meters; signal sources; sensors, LVDT, strain gauge and temperature/flow transducers..

LAST-MILE REVISION

Quick Revision

Module 1: Measurement Fundamentals and Analog Instruments

Measurement terms, static characteristics, errors, PMMC/MI/electrodynamometer/induction/electrostatic instruments and range extension.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Instrument Transformers, Power and Energy Measurement

CT/PT principles and errors; wattmeters; single/three-phase power; induction energy meters and testing.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Potentiometers and Bridge Measurements

DC/AC potentiometers; low/medium/high resistance; Wheatstone/Kelvin bridges; inductance/capacitance/frequency bridges.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Electronic Instruments, Oscilloscopes and Transducers

CRO/DSO operation; voltage/frequency/phase measurement; digital meters; signal sources; sensors, LVDT, strain gauge and temperature/flow transducers.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Measurement systems, characteristics and errors	Accuracy, precision, resolution, sensitivity and speed of response describe measurement quality.
PMMC, moving-iron and electrodynamometer instruments	Analog instruments convert electrical quantity into deflecting torque balanced by control torque.
Range extension and electrostatic instruments	Shunts extend current ranges and series multipliers extend voltage ranges.
Current and potential transformers	Instrument transformers scale high current or voltage to safe standard values for meters and relays.
Wattmeters and power measurement	Electrodynamometer and induction-type principles support power measurement.
Energy meters	An energy meter accumulates electrical energy over time.
Potentiometers	A potentiometer compares an unknown voltage with a known calibrated voltage under null-balance condition, minimising loading of the source.
Resistance measurement and DC bridges	Medium resistance is commonly measured through bridge comparison; Kelvin methods reduce lead/contact influence for low resistance; high-resistance methods use appropriate insulation/leakage precautions.
AC bridges	AC bridges compare impedance to determine inductance, capacitance, resistance or frequency.
CRO and digital storage oscilloscope	Oscilloscopes display voltage versus time and support amplitude, time, frequency and phase interpretation.
Digital instruments and signal generators	Digital voltmeters, multimeters and frequency meters provide numerical readings with specified resolution, sensitivity and input characteristics.
Sensors and transducers	A transducer converts a physical quantity into a usable signal.

LEARNING RESOURCES

References

Recommended electrical-measurement references

1. A. K. Sawhney, Electrical and Electronic Measurements and Instrumentation.
2. E. W. Golding and F. C. Widdis, Electrical Measurements and Measuring Instruments.
3. H. S. Kalsi, Electronic Instrumentation.

Approved public electrical-measurement sources

- https://files.mlrit.ac.in/curriculum/EEE/EEE-MLR20/2-1/Electrical-Measurements-and-Instrumentation.pdf
- https://hithaldia.in/hit_resource/course/ee_course/4/EE402.pdf

These sources support study explanations while the official Course 3031 detail page is unavailable; they do not replace an official detailed syllabus.